



STUDENT NUMBER: _____

TEACHER: _____

THGS Year 12 Maths Ext 1 Task 4 Trial 2014

THE HILLS GRAMMAR SCHOOL

Trial Higher School Certificate Examination 2014

MATHEMATICS EXTENSION 1

Time Allowed: Two hours (plus five minutes reading time)

Weighting: %

Outcomes: H6, H7, H8, H9, HE1, HE2, HE4, HE7, HE9

General Instructions:

- Board-approved calculators may be used
- Attempt all questions
- Start all questions on a new sheet of paper
- The marks for each question are indicated on the examination
- Show all necessary working for Questions 11-14
- The diagrams are not drawn to scale
- A table of standard integrals is provided

Total Marks – 70

Section I Questions 1-10

10 Marks

Allow about 15 minutes for this section

Section II Questions 11-14

60 Marks

Allow about 1 hour and 45 minutes for this section

MCQ	Question 11	Question 12	Question 13	Question 14	TOTAL
10	15	15	15	15	70

Section 1 Multiple Choice (10 Marks)

1 When $2x^3 - 3x^2 + 2a - 4$ is divided by $x - 1$ the remainder is -5. The value of a is:

- (A) 2 (C) -2
(B) 0 (D) -3

2 The domain and range of $f(x) = 3 \sin^{-1}\left(\frac{x}{2}\right)$ is given by:

- (A) x is real
 $-3 \leq y \leq 3$
(B) $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$
 $-3 \leq y \leq 3$
(C) $-\frac{1}{2} \leq x \leq \frac{1}{2}$
 $-\frac{3\pi}{2} \leq y \leq \frac{3\pi}{2}$
(D) $-2 \leq x \leq 2$
 $-\frac{3\pi}{2} \leq y \leq \frac{3\pi}{2}$

3 The angle between $y = 2x + 3$ and $y = x^2$ when $x = 3$ is given by:

- (A) 0° (C) 90°
(B) $\tan^{-1}\left(\frac{4}{13}\right)$ (D) $\tan^{-1}\left(-\frac{8}{11}\right)$

4 If the interval AB is divided externally in the ratio 3:1 by the point P , the coordinates of P given $A(-2, 3)$ and $B(3, -4)$ are:

- (A) $\left(\frac{11}{2}, -\frac{15}{2}\right)$
(B) $\left(-\frac{1}{2}, \frac{1}{2}\right)$
(C) $\left(-\frac{11}{2}, \frac{15}{2}\right)$
(D) $\left(-\frac{1}{2}, -\frac{1}{2}\right)$

5 The equation of the tangent to the parabola $y^2 = 4ax$ at the point $(ap^2, 2ap)$ is given by:

(A) $px - y - ap^2 = 0$ (C) $px + y - ap^2 = 0$

(B) $x - py + ap^2 = 0$ (D) $x - py - ap^2 = 0$

6 The coefficient of x^5 in $\left(x^2 - \frac{2}{x}\right)^7$ is:

(A) ${}^7C_3(-2)^3$ (B) ${}^7C_4(-2)^4$

(C) ${}^7C_5(-2)^5$ (D) ${}^7C_4(-2)^3$

7 Evaluate $\lim_{x \rightarrow 0} \frac{x}{\tan 2x}$:

(A) 0 (C) 2

(B) ∞ (D) 0.5

8 The derivative of $\tan^{-1}\left(\frac{x^3}{3}\right)$ is:

(A) $\frac{3x^2}{9+x^6}$ (C) $\frac{3x^2}{1+x^6}$

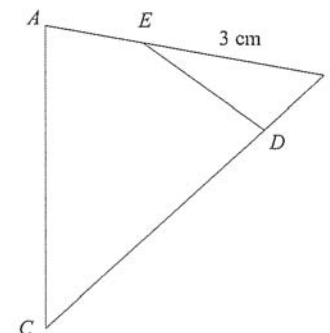
(B) $\frac{x^2}{9+x^6}$ (D) $\frac{9x^2}{9+x^6}$

9 If $t = \tan\left(\frac{\theta}{2}\right)$ the correct expression for $\frac{\sec^2 \theta}{\operatorname{cosec}^2 \theta}$ is:

(A) $\frac{4t^2}{(1-t^2)^2}$ (B) $\frac{(1+t^2)^2}{(1-t^2)^2}$

(C) $\frac{(1+t^2)}{(1-t^2)^2}$ (D) $\frac{(1-t^2)^2}{4t^2}$

10 In the diagram below $BE = 3$ cm, $AE = BD = x$, $DC = 11x$ and $\angle BDE = \angle BAC$.



What is the value of x ?

(A) $\frac{1}{2}$

(B) $\frac{3}{4}$

(C) 1

(D) $1\frac{1}{2}$

Section 2

Marks

Question 11 (15 marks)

(a) Use the substitution $u = 1 + x$ to evaluate $15 \int_{-1}^0 x\sqrt{1+x} \, dx$ 3

(b) Let $f(x) = 3x^2 + x$. Use the definition $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ to find the derivative of $f(x)$ at the point $x = a$. 2

(c) Find

(i) $\int \frac{e^x}{1+e^x} \, dx$ 1

(ii) $\int_0^{\pi} \cos^2 3x \, dx$ 3

(d) Find the term independent of x in the binomial expansion of $\left(x^2 - \frac{1}{x}\right)^9$ 3

(e) By using the binomial expansion,

(i) show that $(q+p)^n - (q-p)^n = 2\binom{n}{1}q^{n-1}p + 2\binom{n}{3}q^{n-3}p^3 + \dots$ 1

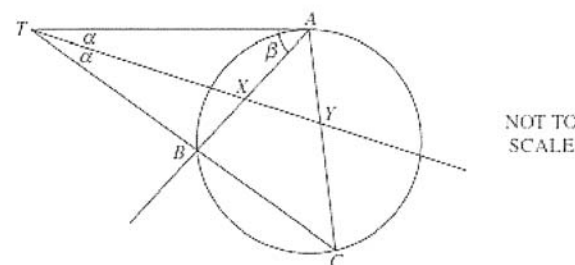
(ii) What is the last term in the expansion if n is odd? 1

(iii) What is the last term in the expansion if n is even? 1

START A NEW PAGE

Question 12 (15 marks)

- (a) In the diagram the points A , B and C lie on the circle and CB produced meets the tangent from A at the point T . The bisector of the angle ATC intersects AB and AC at X and Y respectively. Let $\angle TAB = \beta$.



Copy or trace the diagram into your writing booklet.

- (i) Explain why $\angle ACB = \beta$ 1
- (ii) Hence prove that triangle AXY is isosceles. 2

- (b) A household iron is cooling in a room of constant temperature 22°C . At time t minutes its temperature T decreases according to the equation

$$\frac{dT}{dt} = -k(T - 22) \text{ where } k \text{ is a positive constant.}$$

The initial temperature of the iron is 80°C and it cools to 60°C after 10 minutes.

- (i) Verify that $T = 22 + Ae^{-kt}$ is a solution of this equation, where A is a constant. 1
- (ii) Find the values of A and k . (give answers to 2 significant figures) 2
- (iii) How long will it take for the temperature of the iron to cool to 30°C ? (Give your answer to the nearest minute.) 2

(c) The polynomial $P(x) = x^3 - 2x^2 + kx + 24$ has roots α, β, γ .

(i) Find the value of $\alpha + \beta + \gamma$. 1

(ii) Find the value of $\alpha\beta\gamma$. 1

(iii) It is known that two of the roots are equal in magnitude but opposite in sign. Find the third root and hence find the value of k . 2

(d) Use the principle of mathematical induction to show that

$2 \times 1! + 5 \times 2! + 10 \times 3! + \dots + (n^2 + 1)n! = n(n+1)!$ for all positive integers n . 3

START A NEW PAGE

Question 13 (15 marks)

(a) If $f(x) = \ln(x+3)$

(i) find $f^{-1}(x)$. 1

(ii) Sketch $y = x$, $f(x)$ and $f^{-1}(x)$ on the same axes. 2

(b) A particle moves in a straight line and its position at time t is given by

$$x = 4 \sin\left(2t + \frac{\pi}{3}\right)$$

(i) Show that the particle is undergoing simple harmonic motion. 2

(ii) Find the amplitude of the motion. 1

(iii) When does the particle first reach maximum speed after time $t = 0$? 1

(c) The acceleration of a particle P is given by the equation

$$\frac{d^2x}{dt^2} = 8x(x^2 + 4)$$

where x metres is the displacement of P from a fixed point O after t seconds. Initially the particle is at O and has velocity 8 ms^{-1} in the positive direction.

(i) Show that the speed at any position x is given by $2(x^2 + 4) \text{ ms}^{-1}$. 2

(ii) Hence find the time taken for the particle to travel 2 metres from O . 2

- (d) A particle is projected from the origin with velocity $v \text{ ms}^{-1}$ at an angle α to the horizontal. The position of the particle at time t seconds is given by the parametric equations

$$x = vt \cos \alpha$$

$$y = vt \sin \alpha - \frac{1}{2}gt^2 \quad (\text{Do not prove these equations.})$$

- (i) Show that the maximum height reached, h metres, is given by

$$h = \frac{v^2 \sin^2 \alpha}{2g} \quad 2$$

- (ii) Show that it returns to the initial height at $x = \frac{v^2}{g} \sin 2\alpha$ 2

START A NEW PAGE

Question 14 (15 marks)

- (a) (i) Write $8 \cos x + 6 \sin x$ in the form $A \cos(x - \alpha)$ where $A > 0$ and $0 \leq \alpha \leq \frac{\pi}{2}$, 2

- (ii) Hence, or otherwise, solve the equation $8 \cos x + 6 \sin x = 5$ for $0 \leq x \leq 2\pi$. 2
Give your answers correct to three decimal places.

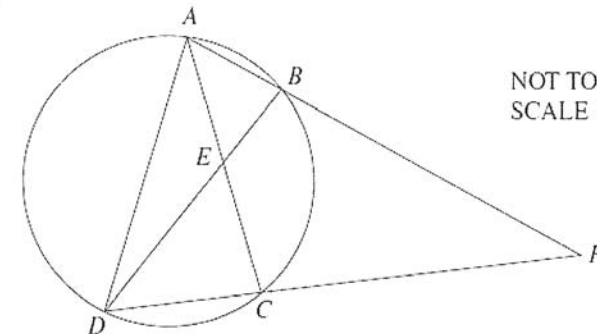
- (b) The two points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ are on the parabola $x^2 = 4ay$.

- (i) The equation of the tangent to $x^2 = 4ay$ ($2at, at^2$) at P is $y = px - ap^2$. (Do not prove this.)

Show that the tangents at the points P and Q meet at R , where R is the point $[a(p+q), apq]$. 2

- (ii) As P varies, the point Q is always chosen so that $\angle POQ$ is a right angle, where O is the origin. Using this condition and the result of part (i) find the locus of R . 2

(c)



The points A, B, C and D are placed on a circle of radius r such that AC and BD meet at E . The lines AB and DC are produced to meet at F , and $BECF$ is a cyclic quadrilateral. Copy or trace this diagram into your writing booklet.

- (i) Find the size of $\angle DBF$, giving reasons for your answer. 2

- (ii) Explain why AD equals $2r$. 1

(d)

(i) Show that for all positive integers n ,

$$x[(1+x)^{n-1} + (1+x)^{n-2} + \dots + (1+x)^2 + (1+x) + 1] = (1+x)^n - 1 \quad 2$$

(ii) Hence explain why

$$\binom{n-1}{k-1} + \binom{n-2}{k-1} + \binom{n-3}{k-1} + \dots + \binom{k-1}{k-1} = \binom{n}{k} \quad \text{for } 1 \leq k \leq n \quad 1$$

$$\text{(iii) Show that } n \binom{n-1}{k} = (k+1) \binom{n}{k+1} \quad 1$$

END OF ASSESSMENT

Trial Exam 2014 Draft 1 (Ext 1)

Q 1, 13
MCG

$$1, \quad f(x) = 2x^3 - 3x^2 + 2x - 4$$

$$f(1) = 2 - 3 + 2 - 4 = -3$$

$$2a - 10 = -5$$

(B) (B)

$$2, \quad f(x) = 3 \sin^{-1} \frac{x}{2}$$

domain $-1 \leq \frac{x}{2} \leq 1$
 $-2 \leq x \leq 2$

range $-3\frac{\pi}{2} \leq y \leq 3\frac{\pi}{2}$

(D)

$$3, \quad y = 2x + 3 \Rightarrow m_1 = 2$$

$$y = x^2$$

$$\frac{dy}{dx} = 2x \quad \text{at } x=3 \quad m_2 = 6.$$

$$\tan \theta = \frac{6-2}{1+6 \cdot 2} = \frac{4}{13}$$

(B)

$$4, \quad A(-2, 3) \rightarrow B(3, -4)$$

3: -1

$$P = \left(\frac{-2+9}{2}, \frac{-3-12}{2} \right) = \left(\frac{11}{2}, -\frac{15}{2} \right)$$

(A)

$$5, \quad y^2 = 4ax$$

$$2y \frac{dy}{dx} = 4a$$

$$\frac{dy}{dx} = \frac{2a}{y} \quad \text{when } y = 2ap.$$

$$\frac{dy}{dx} = \frac{1}{p}$$

equa of tangent $\frac{y - 2ap}{x - ap^2} = \frac{1}{p}$

$$py - 2ap^2 = x - ap^2$$

$$x - py + ap^2 = 0$$

(B)

$$6/ \quad \left(x^2 - \frac{2}{x}\right)^7 \quad T_{r+1} = \binom{7}{r} (x^2)^{7-r} \left(-\frac{2}{x}\right)^r$$

$$= \binom{7}{r} x^{14-2r} \frac{(-2)^r}{x^r} = \binom{7}{r} (-2)^r x^{14-3r}$$

for x^5 term $14-3r=5$
 $-3r = -9 \Rightarrow r = 3$

$$T_4 = \binom{7}{3} (-2)^3$$

(A)

$$7/ \quad \lim_{x \rightarrow 0} \frac{x}{\tan 2x} = \lim_{2x \rightarrow 0} \frac{2x}{\tan 2x} = \frac{1}{2}$$

(D)

$$8/ \quad y = \tan^{-1} \frac{x^3}{3} \quad \frac{dy}{dx} = \frac{x^2}{1 + \frac{x^6}{9}} = \frac{9x^2}{9+x^6}$$

(D)

$$9/ \quad t = \tan\left(\frac{\theta}{2}\right) \quad \frac{\sec^2 \theta}{\csc^2 \theta} = \frac{\sin^2 \theta}{\cos^2 \theta} = \tan^2 \theta = \left(\frac{2t}{1-t^2}\right)^2$$

(A)

$$10/ \quad 2\sin^2 x - \sin x = 0$$

$$\sin x (2\sin x - 1) = 0$$

$$\sin x = 0 \quad \text{or} \quad \sin x = \frac{1}{2}$$

$\downarrow$
 $2x = \pm k\pi$
 $x = \pm \frac{k\pi}{2}$

$\downarrow$
 $2x = \frac{\pi}{6} + 2k\pi \quad \text{or} \quad -\frac{\pi}{6} + (2k+1)\pi$
 $x = \frac{\pi}{12} + k\pi \quad -\frac{\pi}{12} + \frac{(2k+1)\pi}{2}$

(C)

Quest 11

$$(a) \quad 15 \int_{-1}^0 x \sqrt{1+x} \, dx$$

let $u = 1+x$ when $x=-1, u=0$
 $\frac{du}{dx} = 1 \quad x=0, u=1$ (1 mark)

$$I = 15 \int_0^1 (u-1) u^{\frac{1}{2}} du \quad (1 \text{ mark})$$

$$= 15 \int_0^1 u^{\frac{3}{2}} - u^{\frac{1}{2}} du$$

$$= 15 \left[\frac{2}{5} u^{\frac{5}{2}} - \frac{2}{3} u^{\frac{3}{2}} \right]_0^1$$

$$= 15 \left(\frac{2}{5} - \frac{2}{3} \right)$$

$$= 15 \left(\frac{6-10}{15} \right) = -4 \quad (1 \text{ mark})$$

(b) $f(x) = 3x^2 + x$
 $f(a) = 3a^2 + a$
 $f(a+h) = 3(a^2 + 2ah + h^2) + a + h$ (1 mark)

$$\frac{f(a+h) - f(a)}{h} = \frac{6ah + 3h^2 + h}{h}$$

$$= 6a + 3h + 1 \quad (1 \text{ mark})$$

$$f'(a) = \lim_{h \rightarrow 0} (6a + 3h + 1) = 6a + 1$$

(c)(i) $\int \frac{e^x}{1+e^x} dx = \ln(1+e^x) + C$ (1 mark)

(ii) $\int_0^{\pi} \cos^2 3x \, dx = \frac{1}{2} \int_0^{\pi} (\cos 6x + 1) dx$ (1 mark)

$$= \frac{1}{2} \left[\frac{1}{6} \sin 6x + x \right]_0^{\pi} \quad (1 \text{ mark})$$

$$= \frac{\pi}{2}$$

Comments

• Some students failed to convert surd form to indice form.

• Many from 2nd class omitted $\lim_{h \rightarrow 0}$, docked 1 mark.

• Many students mixed terminals
 • could not rewrite $\cos^2 3x$ in terms of $\cos 6x$.

(d) $(x^2 - \frac{1}{x})^9$ (1 mark)

$$T_{r+1} = \binom{9}{r} (x^2)^{9-r} \left(-\frac{1}{x}\right)^r$$

$$= \binom{9}{r} (-1)^r x^{18-2r-r}$$

for const. term $18-3r=0 \Rightarrow r=6$. (1 mark)

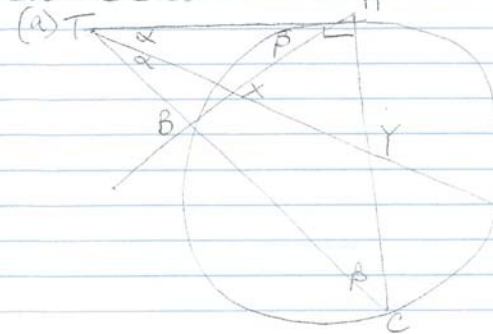
$$T_7 = \binom{9}{6} (-1)^6 x^{18-18} = \binom{9}{6}$$

$$= 84 \quad (1 \text{ mark})$$

Comments

If $(-1)^6$ not shown, 1 mark docked.

Question 12



Comments
 $\angle ACB = \beta$ (angle between chord & tangent equals angle in alternate segment)
 (1 mark)

Students made part (ii) too complicated.

Prove $\triangle AXY$ is isosceles

using $\triangle TAX$ $\angle AXY = \alpha + \beta$ (ext. $\angle$ of $\triangle$) (1 mark)
 $\angle BAX = \angle BCY$

using $\triangle TCY$ $\angle AYX = \alpha + \beta$ (ext. $\angle$ of $\triangle$) (1 mark)
 $\therefore \triangle AXY$ is isosceles

(b) $\frac{dT}{dt} = -k(T-22)$ when $t=0$, $T=80$
 when $t=10$, $T=60$

(i) $T = 22 + Ae^{-kt}$
 $\frac{dT}{dt} = -kAe^{-kt}$
 $\frac{dT}{dt} = -k(T-22)$ (1 mark)

(ii) when $t=0$
 $80 = 22 + Ae^0 \Rightarrow A = 58$ (1 mark)

when $t=10$, $T=60$

$$60 = 22 + 58e^{-k \cdot 10}$$

$$38 = 58e^{-10k}$$

$$\ln \frac{38}{58} = -10k$$

$$k = \ln \frac{38}{58} \div -10$$

$$\approx 0.0423 \Rightarrow \text{mining}$$

Some students got the answer wrong because they did not use the memory key.

(iii) for $T=30^\circ$

$$30 = 22 + 58e^{-kt}$$

$$-kt = \ln\left(\frac{8}{58}\right)$$

$$t = \ln\left(\frac{8}{58}\right) \div -k$$

$$= 46.85$$

$$\approx 47 \text{ mins} \quad (1 \text{ mark})$$

show
 (e) $(q+p)^n - (q-p)^n = 2\binom{n}{1}q^{n-1}p + 2\binom{n}{3}q^{n-3}p^3$

LHS = $q^n + \binom{n}{1}q^{n-1}p + \binom{n}{2}q^{n-2}p^2 + \binom{n}{3}q^{n-3}p^3$
 $- (q^n - \binom{n}{1}q^{n-1}p + \binom{n}{2}q^{n-2}p^2 - \binom{n}{3}q^{n-3}p^3)$ (1 mark)
 $= 2\binom{n}{1}q^{n-1}p + 2\binom{n}{3}q^{n-3}p^3 + 2\binom{n}{5}q^{n-5}p^5 + \dots$

If n is odd last term $2p^n$

If n is even last term $2\binom{n}{n-1}q^{n-1}p$

(1 mark) many swapped ans for odd & even.

(c) $P(x) = x^3 - 2x^2 + kx + 24$
has roots α, β, γ

(i) $\alpha + \beta + \gamma = 2$ ① mark

(ii) $\alpha\beta\gamma = -24$ ① mark

(iii) let roots be $x, -x, \beta$

$$\begin{aligned}\beta &= 2 \\ -x^2\beta &= -24 \\ x &= 12 \\ \alpha &= \pm 2\sqrt{3}\end{aligned}$$

① mark

$$\begin{aligned}k &= \alpha\beta - \alpha\beta + \alpha x - x \\ &= -12\end{aligned}$$

① mark.

(d) Show $2 \times 1! + 5 \times 2! + 10 \times 3! + \dots + (n^2 + 1)n! = n(n+1)!$

Test $n=1$ LHS = $2 \times 1! = 2$

RHS = $1(1+1) = 2$

$\therefore$ True for $n=1$ ① mark

Assume true for $n=k$

$$2 \times 1! + 5 \times 2! + \dots + (k^2 + 1)k! = k(k+1)! \quad * S_k$$

① mark

Test $n=k+1$

$$1 \times 2 \times 1! + 5 \times 2! + \dots + (k^2 + 1)k! + [(k+1)^2 + 1](k+1)! = (k+1)(k+2)! \quad * S_{k+1}$$

$$\text{LHS} = k(k+1)! + [(k+1)^2 + 1](k+1)!$$

$$= (k+1)! [k + k^2 + 2k + 1 + 1]$$

$$= (k+1)! [k^2 + 3k + 2]$$

$$= (k+1)! (k+2)(k+1)$$

$$= (k+1)(k+2)! = \text{RHS}$$

$\therefore$ If $n=k$ true then $n=k+1$ is true

It is true for $n=1$

$\therefore$ True for all $n \in \mathbb{Z}^+$

Comments

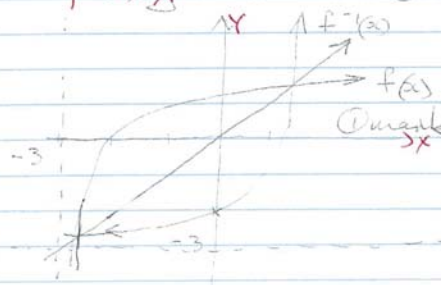
Question 13

(a) $f(x) = \ln(x+3)$

$f^{-1}(x)$ is $x = \ln(y+3)$

$e^x = y+3$

$f^{-1}(x) = e^x - 3$ ① mark



• if not written as $f^{-1}(x)$, no mark awarded.

• many did not sketch properly. labels of graphs axes omitted

(b)(i) $x = 4 \sin\left(2t + \frac{\pi}{3}\right)$

$\dot{x} = 8 \cos\left(2t + \frac{\pi}{3}\right)$

$\ddot{x} = -16 \sin\left(2t + \frac{\pi}{3}\right)$

$\ddot{x} = -4x$ This is SHM

(ii) amplitude = 4 ① mark

(iii) for max speed,

$\cos\left(2t + \frac{\pi}{3}\right) = 1$ or $\sin\left(2t + \frac{\pi}{3}\right) = 0$

$2t + \frac{\pi}{3} = 0, \pi$ etc

$2t = \frac{2\pi}{3}$

$t = \frac{\pi}{3}$ secs ① mark.

• students failed to write -16 as $-4x$

• ans poorly.

Comments

(c) $\frac{d^2x}{dt^2} = 8x(x^2+4)$ when $t=0, x=0, v=8$

$$\frac{d}{dt} = 8x^3 + 32x$$

$$\frac{d}{dt} \left(\frac{1}{2} v^2 \right) = 8x^3 + 32x$$

$$\frac{1}{2} v^2 = 2x^4 + 16x^2 + c \quad \text{1 mark}$$

$$v^2 = 4x^4 + 32x^2 + 2c$$

$$\text{when } x=0, v=8 \Rightarrow 2c=64$$

$$v^2 = 4x^4 + 32x^2 + 64$$

$$v^2 = 4(x^4 + 8x^2 + 16)$$

$$v = \pm 2(px^2 + 4) \text{ ms}^{-1}$$

when particle commences it

has $v > 0$ and $\dot{x} > 0$

$$\therefore v = 2(x^2 + 4) \text{ ms}^{-1} \quad \text{1 mark}$$

(ii) $\frac{dx}{dt} = 2(x^2 + 4)$

$$\frac{dt}{dx} = \frac{1}{2(x^2 + 4)} \quad \text{1 mark}$$

$$t = \frac{1}{2} \times \left[\frac{1}{2} \tan^{-1} \frac{x}{2} \right]_0^x$$

$$t = \frac{1}{4} \tan^{-1} \frac{x}{2}$$

$$t = \frac{\pi}{16} \text{ sec} \quad \text{1 mark}$$

(d) $x = vt \cos \alpha$

$$y = vt \sin \alpha - \frac{1}{2} g t^2$$

(i) $y = v \sin \alpha - g t$

for max height $y=0$

$$g t = v \sin \alpha$$

$$t = \frac{v \sin \alpha}{g} \quad \text{1 mark}$$

$$h = v \sin \alpha \frac{v \sin \alpha}{g} - \frac{1}{2} g \frac{v^2 \sin^2 \alpha}{g^2}$$

$$= \frac{v^2 \sin^2 \alpha}{g} - \frac{1}{2} \frac{v^2 \sin^2 \alpha}{g}$$

$$= \frac{v^2 \sin^2 \alpha}{2g}$$

• poorly ans.
• students
failed to calc
c
• lots of fudging
in c + d.

• poorly answered

• students did
not get to $y=0$
for max h.

• 1 Mark
awarded
for $t = \frac{v \sin \alpha}{g}$
• 2 Marks
given for attempt
to subst in y.

Comments

(ii) $x = vt \cos \alpha$ ①, $y = vt \sin \alpha - \frac{1}{2} g t^2$ ②

$$t = \frac{x}{v \cos \alpha} \quad \text{③}$$

sub ③ in ②

$$y = x \tan \alpha - \frac{g}{2} \frac{x^2}{v^2 \cos^2 \alpha}$$

$$y = x \tan \alpha - \frac{g}{2} \frac{x^2}{v^2 \cos^2 \alpha}$$

for $y=0$

$$x \tan \alpha = \frac{g}{2} \frac{x^2}{v^2 \cos^2 \alpha} \quad \text{1 mark}$$

$x=0$ at start for $x \neq 0$

$$\frac{g}{2v^2 \cos^2 \alpha} = \tan \alpha$$

$$\alpha = \frac{2v^2 \cos^2 \alpha}{g} \frac{\sin \alpha}{\cos \alpha}$$

$$\alpha = \frac{v^2 \sin 2\alpha}{g} \quad \text{1 mark}$$

Question 14

a) i) let $8 \cos \alpha + 6 \sin \alpha = A \cos(\alpha - \alpha)$

$$= A \cos \alpha \cos \alpha + A \sin \alpha \sin \alpha$$

$$A \cos \alpha = 8 \quad \alpha \text{ in 1st quad}$$

$$A \sin \alpha = 6$$

$$\tan \alpha = \frac{6}{8} = \frac{3}{4}$$

$$\alpha = \tan^{-1} \left(\frac{3}{4} \right) \quad \text{1 mark}$$

$$A^2 = 6^2 + 8^2$$

$$A^2 = 100$$

$$A = 10$$

1 mark

$$8 \cos \alpha + 6 \sin \alpha = 10 \cos(\alpha - \alpha)$$

ii) $8 \cos \alpha + 6 \sin \alpha = 5$

$$10 \cos(\alpha - \alpha) = 5$$

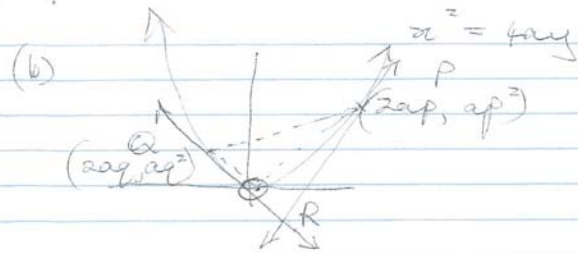
$$\cos(\alpha - \alpha) = \frac{1}{2} \quad \text{1 mark}$$

$$\alpha - \alpha = \frac{\pi}{3}, 2\pi - \frac{\pi}{3}$$

1 mark

$$\alpha = \frac{\pi}{3} + \tan^{-1} \left(\frac{3}{4} \right), \frac{5\pi}{3} - \tan^{-1} \left(\frac{3}{4} \right)$$

• poorly answered
• lots of
fudging.



tangent thru P is $y = px - ap^2$ (1)
 " thru Q is $y = qx - aq^2$ (2)
 equate (1) & (2)

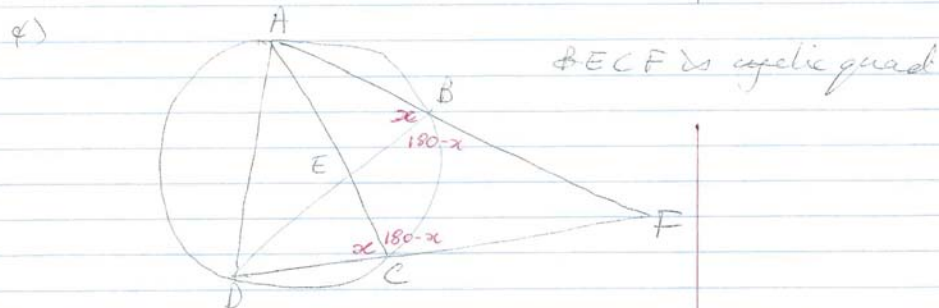
$$px - ap^2 = qx - aq^2$$

$$(p - q)x = ap^2 - aq^2$$

$$x = a(p + q) \quad (1 \text{ mark})$$

sub in (1) $y = ap(p + q) - ap^2$
 $= apq \quad (1 \text{ mark})$
 $\therefore R = (a(p + q), apq)$

(ii) for $\angle POQ = 90^\circ$ ~~pg = -1~~ grad of OP = $\frac{ap^2}{2ap} = \frac{p}{2}$
 $\therefore pq = -1$ gradients (1 mark) $pg = -4$
 $\therefore x = \frac{4}{p} a(p + q)$
 $y = -4a \leftarrow \text{locus}$
 ~~$y = -4a(p + q)$~~
 $\therefore \text{locus is line } y = -4a \quad (1 \text{ mark})$
 i.e. directrix



(i) let $\angle ABD = x$ (1 mark)
 then $\angle ACD = x$ (angles on same arc AD)
 $\therefore \angle EBF = \angle ECF$ (supp. $\angle$'s on line)
 $\angle EBF + \angle ECF = 180^\circ$ (opp $\angle$'s in cyclic quad)
 $180 - x + 180 - x = 180 \therefore x = 90^\circ \quad (1 \text{ mark})$

Comments

(i) If $\angle ABD = 90^\circ$
 then AD is a diameter ($\triangle ADB$ is right angled)
 hence $AD = 2a \quad (1 \text{ mark})$

(d) Show $x[(1+x)^{n-1} + (1+x)^{n-2} + \dots + (1+x)^2 + (1+x) + 1] = (1+x)^n - 1$

(i) LHS = $x \times \text{GP}$ with $a = 1$, $r = (1+x)$, n terms.

$$= x \left(\frac{a(1+x)^n - 1}{(1+x) - 1} \right) \quad (1 \text{ mark})$$

$$= (1+x)^n - 1$$

$$= \text{RHS} \quad (1 \text{ mark})$$

Some students used Induction.

(ii) x^k term on RHS = $\binom{n}{k}$

$$x^k \text{ term on LHS} = \binom{n-1}{k-1} + \binom{n-2}{k-1} + \dots + \binom{k-1}{k-1}$$

(iii) Show that $n \binom{n-1}{k} = (k+1) \binom{n}{k+1}$

$$\text{LHS} = \frac{n(n-1)!}{k!(n-1-k)!}$$

$$= \frac{n!}{k!(n-1-k)!}$$

$$= \frac{(k+1)n!}{(k+1)k!(n-(k+1))!}$$

$$= \frac{(k+1)n!}{(k+1)!(n-(k+1))!}$$

$$= (k+1) \binom{n}{k+1}$$